

CUSTOMER BEHAVIOR ANALYSIS

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Course: B.Sc. Data Analytics
Tools: Python, SQL, Power BI
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Project Overview

This project focuses on analyzing customer purchasing behavior using data analytics techniques. The objective is to identify patterns, segment customers, and generate insights to support business decision-making. Tools like Python, SQL, and Power BI were used.

Introduction

Customer behavior analysis plays a crucial role in understanding purchasing patterns, preferences, and trends. Businesses use this analysis to improve decision-making, optimize marketing strategies, and enhance customer experience. Using tools like Python, SQL, and Power BI, raw data was transformed into meaningful insights

Objective

- To analyze customer purchasing behaviour and identify patterns in buying habits
- To segment customers based on demographic factors such as age and gender, as well as behavioural factors like purchase frequency
- To identify high-value and high-frequency customers who contribute significantly to revenue
- To evaluate product performance based on sales and customer ratings
- To perform data cleaning and preprocessing to ensure data accuracy and consistency
- To apply feature engineering techniques for better data representation and analysis
- To use SQL queries to extract meaningful business insights from the dataset
- To develop an interactive dashboard using Power BI for effective data visualization
- To support business decision-making through data-driven insights

Dataset Description:

The dataset consists of **3,900 records and 18 features**, containing customer demographic and transactional information.

Key attributes include:

Customer ID, Age, Gender, Item Purchased, Category, Purchase Amount (USD), Location, Size, Colour, Season, Review Rating, Subscription Status, Shipping Type, Discount Applied, Promo Code Used, Previous Purchases, Payment Method, and Frequency of Purchases.

The dataset contains a mix of **numerical and categorical variables**. Most columns have complete data, while the **Review Rating column contains some missing values**, which were handled during preprocessing.

Categorical variables such as **Frequency of Purchases** were converted into numerical form for analysis, and data quality was ensured through cleaning and validation steps.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	IV
freq	NaN	NaN	2652	171	1737	NaN	96	1755
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN

Data Cleaning and Preparation

A structured data preparation and cleaning process was performed using Python to ensure the dataset was accurate, consistent, and suitable for analysis. The preprocessing workflow included data exploration, handling missing values, feature engineering, data standardization, and database integration.

1. Data Loading & Initial Exploration

The dataset was imported into Python using the **Pandas** library for efficient data handling and manipulation.

Initial exploration was conducted to understand the structure and quality of the dataset using functions such as:

- `df.head()` to preview the first few records
- `df.describe(include='all')` to generate summary statistics

- `df.info()` to examine data types and non-null values
- `df.isnull().sum()` to identify missing values

This step helped in understanding the distribution of variables, identifying missing data, and detecting inconsistencies within the dataset.

```
In [4]: import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')
```

```
In [5]: df.head()
```

```
Out[5]:
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	

```
In [6]: df.describe(include= 'all')
```

```
Out[6]:
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900

```
In [7]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                ---
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   object
3   Item Purchased                      3900 non-null   object
4   Category                            3900 non-null   object
5   Purchase Amount (USD)               3900 non-null   int64
6   Location                            3900 non-null   object
7   Size                                3900 non-null   object
8   Color                               3900 non-null   object
9   Season                              3900 non-null   object
10  Review Rating                       3863 non-null   float64
11  Subscription Status                 3900 non-null   object
12  Shipping Type                      3900 non-null   object
13  Discount Applied                   3900 non-null   object
14  Promo Code Used                    3900 non-null   object
15  Previous Purchases                  3900 non-null   int64
16  Payment Method                     3900 non-null   object
17  Frequency of Purchases              3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
In [8]: df.isnull().sum()
```

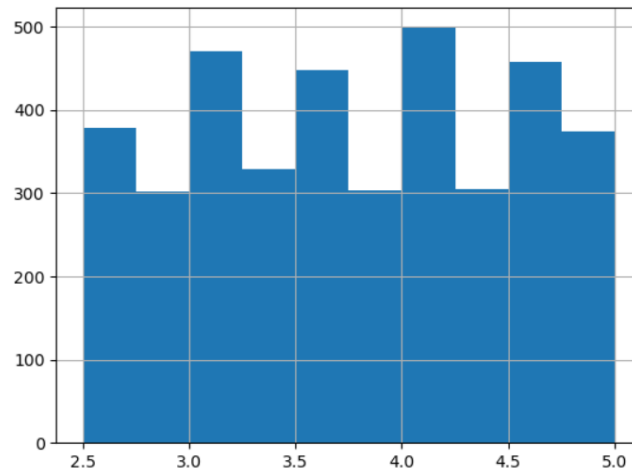
```
Out[8]: Customer ID      0
Age      0
Gender    0
Item Purchased  0
Category  0
Purchase Amount (USD)  0
Location  0
Size      0
Color     0
Season    0
Review Rating  37
Subscription Status  0
Shipping Type  0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method  0
Frequency of Purchases  0
dtype: int64
```

2. Handling Missing Values

- Missing values were identified in the **Review Rating** column during the initial data assessment.
- Before handling missing values, the distribution and skewness of the column were analyzed to better understand the nature of the data.
- Missing values were treated using **category-wise median imputation**, where missing ratings were replaced with the median rating of their respective product category.
- This approach ensured that the imputed values remained contextually relevant and minimized the impact of missing data on overall analysis.
- After imputation, the dataset was revalidated to confirm that missing values had been handled successfully.

```
In [9]: df['Review Rating'].hist()
```

```
Out[9]: <Axes: >
```



```
In [10]: df['Review Rating'].skew()
```

```
Out[10]: 0.0013552919012980596
```

```
In [11]: df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

```
In [12]: df.isnull().sum()
```

```
Out[12]: Customer ID      0
Age                    0
Gender                 0
Item Purchased         0
Category               0
Purchase Amount (USD)  0
Location              0
Size                  0
Color                  0
Season                 0
Review Rating          0
Subscription Status    0
Shipping Type          0
Discount Applied       0
Promo Code Used        0
```

3. Duplicate Detection

- The dataset was checked for duplicate records to ensure data consistency and reliability.
- Duplicate detection was performed as part of the data validation process to verify that no unnecessary repeated transactions existed within the dataset.
- This step helped ensure the accuracy of further analysis and business insights.

4. Column Standardization

- Column names were standardized to improve readability and maintain consistency throughout the analysis process.

- Spaces in column names were replaced with underscores, and all column names were converted into lowercase format.
- Certain lengthy column names were also renamed for simplicity and easier querying during SQL analysis.
- Standardization helped maintain uniform naming conventions across Python, SQL, and Power BI environments.

```
In [13]: df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

5. Data Type Validation

- Data types were verified to ensure correct classification of variables into numerical and categorical formats.
- Numerical columns such as Age, Purchase Amount, and Previous Purchases were validated for quantitative analysis, while categorical columns such as Gender, Category, and Payment Method were checked for consistency.
- Proper data type validation ensured accurate calculations, aggregations, and visualizations during later stages of the project.

6. Feature Engineering

Feature engineering was performed to create additional meaningful variables that improved the analytical capability of the dataset.

► Age Group Creation

- Customers were segmented into categories such as Young Adult, Adult, Middle-aged, and Senior based on their age values.
- This segmentation enabled demographic-based analysis and helped identify purchasing trends across different age groups.

► Purchase Frequency Transformation

- The **Frequency of Purchases** column originally contained categorical values such as *Weekly*, *Monthly*, and *Fortnightly*.
- These categories were converted into numerical equivalents representing the number of days between purchases.

- This transformation enabled quantitative analysis of customer purchasing behavior and simplified comparison across customers.

```
In [16]: #Feature Engineering
labels = ['Young Adult','Adult','Middle_aged','Senior']
df['age_group'] = pd.qcut(df['age'],q=4,labels = labels)
```

```
In [17]: df[['age', 'age_group']].head(10)
```

```
Out[17]:
```

	age	age_group
0	55	Middle_aged
1	19	Young Adult
2	50	Middle_aged
3	21	Young Adult
4	45	Middle_aged
5	46	Middle_aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle_aged

```
In [18]: # Create column purchase_frequency_days

frequency_mapping = {
    'Fortnightly' : 14,
    'Weekly' : 7,
    'Monthly' : 30,
    'Quarterly': 90,
    'Bi-Weekly' : 14,
    'Annually' : 365,
    'Every 3 Months': 90
}

df['Purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```



```
In [19]: df[['Purchase_frequency_days', 'frequency_of_purchases']].head(10)
```

```
Out[19]:
```

	Purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

7. Data Consistency and Redundancy Checks

- The dataset was examined for redundant and overlapping information to improve efficiency and reduce unnecessary complexity.
- A consistency check was performed between the columns **Discount Applied** and **Promo Code Used** to verify whether both represented similar information.
- Since both columns contained identical values, the redundant column was removed from the dataset.
- These checks ensured that the dataset remained clean, optimized, and suitable for accurate analysis.

```
In [20]: df[['discount_applied', 'promo_code_used']].head(10)
```

```
Out[20]:
```

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

```
In [21]: (df['discount_applied'] == df['promo_code_used']).all()
```

```
Out[21]: True
```

```
In [22]: df = df.drop('promo_code_used', axis=1)
```

8. Database Integration

- After preprocessing, the cleaned dataset was integrated into **PostgreSQL** using **SQLAlchemy**.
- The processed DataFrame was loaded into a PostgreSQL table to enable efficient SQL-based querying and business analysis.
- Database integration improved data accessibility and supported advanced analytical operations.

```
n [3]: pip install psycopg2-binary sqlalchemy
```

```
Requirement already satisfied: psycopg2-binary in c:\users\hp\anaconda3\lib\site-packages (2.9.11)  
Requirement already satisfied: sqlalchemy in c:\users\hp\anaconda3\lib\site-packages (1.4.46)  
Requirement already satisfied: greenlet!=0.4.17 in c:\users\hp\anaconda3\lib\site-packages (from sqlalchemy) (2.0.1)  
Note: you may need to restart the kernel to use updated packages.
```

```
[27]: from sqlalchemy import create_engine  
username = "postgres"  
password = "angel%40123%23"  
host = "localhost"  
port = "5432"  
database = "customer_behavior"  
  
engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")  
  
# Load dataframe into PostgreSQL  
table_name = "customer"  
  
df.to_sql(table_name, engine, if_exists="replace", index=False)  
  
print(f"Data successfully loaded into table '{table_name}' in database '{database}'.")
```

```
Data successfully loaded into table 'customer' in database 'customer_behavior'.
```

Data Analysis using SQL

Structured SQL queries were performed in PostgreSQL to extract meaningful business insights from the cleaned dataset. The analysis focused on customer behavior, product performance, revenue trends, and customer segmentation.

1. Revenue Analysis by Gender

Objective

To analyze the total revenue contribution of male and female customers.

SQL Query

```
-- Q1. What is the total revenue generated by male vs. female customers?
select gender, sum(purchase_amount) as revenue
from customer
group by gender;
```

Output

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Interpretation

Male customers contributed significantly higher revenue (**157,890**) compared to female customers (**75,191**), indicating stronger purchasing contribution within the dataset.

2. Top 5 Products with Highest Average Ratings

Objective

To identify products with the highest customer satisfaction based on average review ratings.

SQL Query

```
-- Q3. Which are the top 5 products with the highest average review rating?
select item_purchased , Round(avg(review_rating)::numeric, 2) as "Average Product Rating"
from customer
group by item_purchased
order by avg(review_rating) desc
limit 5 ;
```

Output:

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Interpretation:

The analysis identified **Gloves, Sandals, Boots, Hat, and Skirt** as the top five highest-rated products based on average customer reviews. **Gloves received the highest average rating (3.86)**, followed closely by other products with ratings ranging between **3.78 and 3.84**. Since the rating differences are relatively small, it indicates a **consistent level of customer satisfaction across these products**.

3. Subscriber vs Non-Subscriber Spending Analysis

Objective

To compare customer spending behavior between subscribed and non-subscribed users.

SQL Query

```
-- Q5. Do subscribed customers spend more? Compare average spend and total revenue between
--subscribers and non-subscribers.
select subscription_status,
count(customer_id) as total_customers,
round(avg(purchase_amount):: numeric,2) as avg_spend,
round(sum(purchase_amount)::numeric,2) as total_revenue
from customer
group by subscription_status
order by total_revenue desc,avg_spend desc;
```

Output:

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	No	2847	59.87	170436.00
2	Yes	1053	59.49	62645.00

Interpretation:

The analysis revealed that **most customers (2,847)** did not have a subscription, while **1,053 customers were subscribed**. Non-subscribed customers generated a significantly higher **total revenue (170,436)** compared to subscribed customers (**62,645**), mainly due to their larger customer base. However, the **average spending between both groups is relatively similar**, indicating that subscription status had minimal impact on individual spending behaviour within the dataset.

4. Customer Segmentation Analysis

Objective

To classify customers into behavioural segments based on purchase history.

SQL Query

```
with customer_type as (
select customer_id , previous_purchases,
case
    when previous_purchases = 1 then 'New'
    when previous_purchases between 2 and 10 then 'Returning'
    else 'Loyal'
end as customer_segment
from customer
)

select customer_segment, count(*) as "Number of Customers"
from customer_type
group by customer_segment;
```

Output:

	customer_segment 	Number of Customers 
1	Loyal	3116
2	New	83
3	Returning	701

Interpretation:

The analysis revealed that **the majority of customers (3,116) are loyal customers**, indicating strong customer retention and repeated purchasing behavior. A moderate number of customers (**701**) fall into the **returning customer segment**, while only **83 customers are new**, suggesting a relatively low acquisition of new customers within the dataset. This indicates strong customer loyalty but also highlights the opportunity to improve strategies for attracting new customers.

5. Top 3 Products within Each Category

Objective

To identify the most frequently purchased products in each product category.

SQL Query

```
-- Q8. What are the top 3 most purchased products within each category?
with item_counts as (
select category, item_purchased,
count(customer_id) as total_orders,
Row_Number() over (partition by category order by count(customer_id) DESC) as item_rank
from customer
group by category, item_purchased
)

select item_rank , category, item_purchased, total_orders
from item_counts
where item_rank <= 3;
|
)
```

Output:

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145

Interpretation:

The analysis identified the **top-performing products within each category based on purchase frequency**. In the **Accessories** category, **Jewelry** recorded the highest number of purchases (**171 orders**), while **Blouse and Pants** emerged as the most purchased products in the **Clothing** category (**171 orders each**). For **Footwear**, **Sandals** were the most frequently purchased product (**160 orders**). These insights help in understanding customer preferences and can support better inventory planning and targeted product promotions.

6. Revenue Contribution by Age Group

Objective

To identify which customer age groups, contribute the most revenue.

SQL Query

```
select age_group ,
sum(purchase_amount) as total_revenue
from customer
group by age_group
order by total_revenue desc;
```

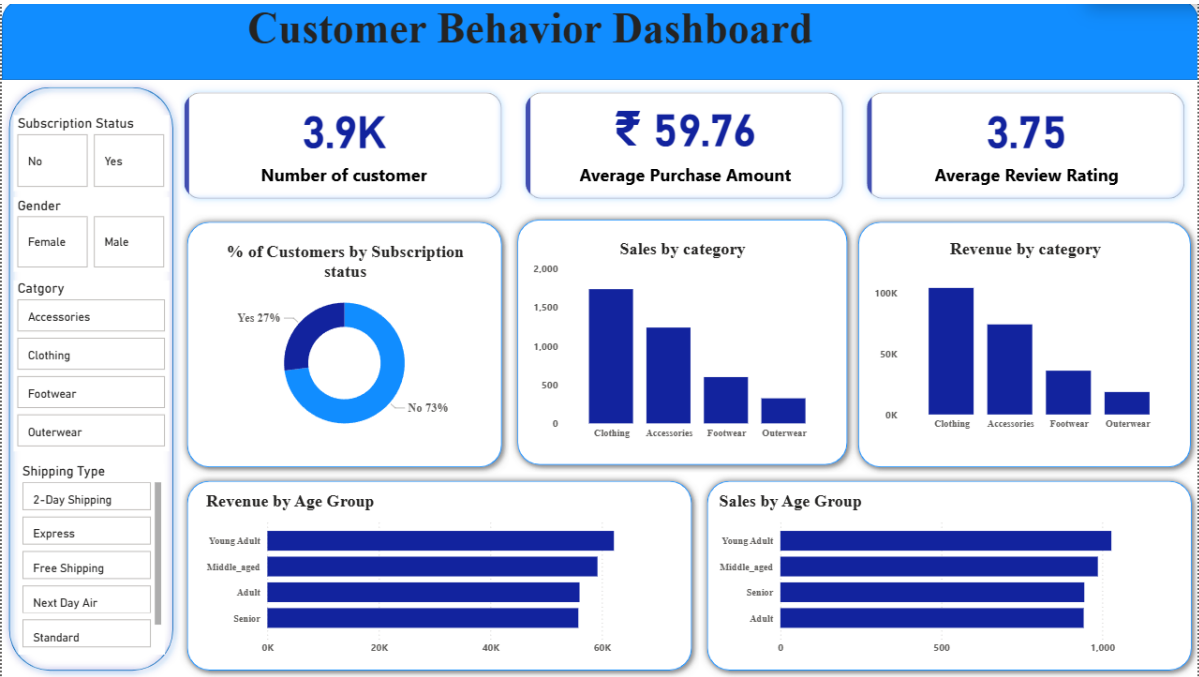
Output:

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle_aged	59197
3	Adult	55978
4	Senior	55763

Interpretation:

The analysis revealed that **Young Adults contributed the highest revenue (62,143)** among all age groups, indicating stronger purchasing activity within this segment. **Middle-aged customers** generated the second-highest revenue (**59,197**), followed closely by **Adult (55,978)** and **Seniors (55,763)**. The relatively balanced revenue distribution suggests that all age groups contribute significantly to overall sales, although **Young Adults represent the most valuable customer segment in terms of revenue generation.**

Dashboard in Power BI :



Business Recommendations:

- Focus marketing efforts on **male and young adult customers**, as they contribute the highest revenue.
- Strengthen **customer retention programs** by offering loyalty rewards and personalized discounts to loyal customers.
- Improve **new customer acquisition** through referral programs, first-time discounts, and social media campaigns.
- Promote **top-rated and high-selling products** such as Gloves, Sandals, and Clothing items to increase sales performance.
- Enhance the **subscription program** by providing exclusive benefits and offers to encourage more customer subscriptions.
- Maintain adequate inventory for **high-performing categories like Clothing** to avoid stock shortages and maximize revenue opportunities.